

Objective and Subjective Metrics Meant for Evaluating Quality of Social Web Applications

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ABSTRACT

The lifetime of majority social Web applications is very short and it is common that developers change their purpose, abandon or terminate project when they realize that it has not met users' expectations. We strongly believe this is due to the lack of suitable methodologies, models, and measuring instruments meant for monitoring and examining quality of social Web applications during their life cycle. To address this issue, we initiated a research on development of a comprehensive methodology that would enable evaluation of both pragmatic and hedonic facets of quality with respect to social Web applications and facilitate their comparison at all levels in the quality model. The work this paper is dealing with presents one of the essential parts of the aforementioned methodology. After establishing the quality model in the form of the requirement tree, objective and subjective metrics in the form of performance variables were introduced. To wrap things up, we proposed elementary criteria defined as preference scales and on the example of two types of social Web applications we illustrated how elementary preference score related to estimated and perceived facets of quality can be determined.

CCS Concepts

• **Software and its engineering** → **Software creation and management** → **Software verification and validation** → **Empirical software validation**

Keywords

Social Web Applications; Quality Evaluation; Objective and Subjective Metrics; Pragmatic and Hedonic Attributes; Logic Scoring of Preference

1. INTRODUCTION

Social Web applications are specific type of software that simplifies users' everyday activities. They enable users to communicate and collaborate in synchronous fashion as well as to create, share, organize, and integrate various artefacts. Some of the most common representatives of social Web applications are wikis, blogs, microblogs, social bookmarking sites, social networking sites, mashups, podcasting applications, e-portfolios, virtual worlds, online office suites, and knowledge management

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applications [1]. Due to their features, social Web applications are widely used for personal, educational and business purposes.

Quality refers to the “degree to which a set of inherent characteristics fulfills a set of requirements” [2]. In order to avoid negative consequences for stakeholders and provide added value, it is essential for every piece of software to be of the high quality [3]. Quality evaluation is therefore considered as one of the essential stages in a software lifecycle. Nowadays, when the development of any kind of software is guided by one of the agile development methodologies, quality evaluation has to be carried out frequently and effectively. Although social Web applications have been used in various aspects of human endeavor for more than a decade, there is still a lack of models, methods, and methodologies that are taking into account features of social Web applications which makes them different from other breeds of software and current evaluation approaches unsuitable in that respect. Our aim is to develop an evaluation methodology that would, based on pragmatic and hedonic quality attributes examined with objective and subjective measuring instruments, enable comparison of social Web applications at all levels of granularity in the evaluation model as well as facilitate determining to what extent user requirements related to a particular quality attribute are met in order to recommend the adequate course of action. One of the first steps towards this goal is to identify objective and subjective metrics that will be used for the evaluation purpose which is what this paper is focused on.

2. RELATED WORK

Considering the subject of the evaluation, current approaches can be, in the context of Web engineering, divided into four groups. The first one are those meant for the assessment of Web sites. In that context, Palmer [4] proposed a set of Web site usability, design, and performance metrics including those related to download delay (initial access speed, speed of display between pages, Alexa speed), navigation/organization (arrangement, sequence, links, layout, Alexa organization), interactivity (customization, interactivity), responsiveness (feedback, FAQ), information/content (amount of information, variety of information, word count, content quality), and Web site success (satisfaction, likelihood of return, frequency of use). Mich et al. [5] introduced 2QCV3Q model which aims to evaluate Web site quality from both owner and user viewpoint through following dimensions: identity (identification, characterization), content (coverage, accuracy), services (functionalities, control), location (reachability, interactivity), management (currentness, maintenance), usability (accessibility, navigability, understandability), feasibility (resources, information and communication technology). Drawing on exhaustive literature review, Hasan and Abuelrub [6] proposed a framework for assessing quality of Web sites that consist of following dimensions:

content quality (timely, relevant, multilanguage/culture, variety of presentation, accuracy, objective, authority), design quality (attractive, appropriateness, color, image/sound/video, text), organization quality (index, mapping, consistency, links, logo), and user-friendly quality (usability, reliability, interactive features, security/privacy, customization).

Next are approaches proposed for examining Web applications. In that respect, Oztekin et al. [7] proposed UWIS, a methodology for measuring usability of web-based information systems composed of following dimensions: efficiency, effectiveness, satisfaction, reliability, integration of communication, navigation, controllability, assurance, responsiveness, and quality of information. Zhao and Zhu [8] developed WebQM, the web quality model focused of following quality aspects: Web source quality (availability, accessibility, durability, and timeliness of information), Web information quality (reliability, correctness, objectivity, understandability, and validity of information), and Web application-specific quality (relevance, presentation, information acquisition).

The third group are approaches for exploring quality of social Web applications in general. Majority of current studies have focused on assessing data quality of social Web applications. For instance, Han [9] measured data quality with respect to Web 2.0 applications with following characteristics: accuracy, completeness, consistency, credibility, currentness, accessibility, compliance, confidentiality, efficiency, precision, traceability, understandability, availability, portability, and recoverability. On the other hand, Pang et al. [10] presented a novel model that examines following faces of quality in the context of Web 2.0 applications: emotional quality (assurance, empathy, interaction, playfulness, and emotion), information quality (completeness, timeliness, comprehensibility, trustworthy, presentation variability, architecture, and search capability), interface quality (proximity, compatibility, navigation, appearance, and layout), service quality (customization, support, channel diversity, responsiveness, incentive, and compensation), and system quality (availability, efficiency, reliability, and security).

Last group refers to approaches introduced for evaluating specific types of social Web applications such as mashups. In that respect, Cappiello et al. [11] proposed a model which includes: data quality (accuracy, timeliness, completeness, availability, and consistency), presentation quality (usability and accessibility), and composition quality (added value, component suitability, component usage, consistency, and availability). As a follow up, Orehovalčki et al. [12] empirically examined model in which following facets of quality with respect to mashups were measured: system quality (efficiency, effectiveness, response time, and compatibility), service quality (availability, reliability, and feedback), content quality (accuracy, completeness, credibility, timeliness, and added value), composition quality (component suitability, composition added value, effectiveness of integrated visualization), effort (minimal memory load, accessibility, ease of use, learnability, and understandability), and user experience (usefulness, playfulness, satisfaction, and loyalty).

3. LOGIC SCORING OF PREFERENCE

Logic scoring of preference [13] (LSP) is a quantitative evaluation method based on continuous preference logic [14] that represents generalization of decision-making techniques. The application of the LSP method consists of three steps [15]. The first one is focused on development of a requirement tree where quality attributes are decomposed to quality indicators and items that can be easily

measured directly. In the context of LSP, quality indicators and items are called performance variables and are denoted as $X_1, X_2, \dots, X_n$ (1). The values of performance variables are commonly real numbers [15]. The second step deals with defining an elementary criterion that is a function $G_i: [X_i^{min}, X_i^{max}] \rightarrow [0,1]$ (2) that for each value of performance variable X_i generates corresponding elementary preference $E_i = G_i(X_i)$, $X_i^{min} \leq X_i \leq X_i^{max}$, $0 \leq E_i \leq 1$, $i = 1, \dots, n$ (3). The elementary criterion is the range of values that a particular performance variable can take and is expressed through a preference scale. The value of elementary preference is normalized between 0 and 1 (or 100%), and represents the degree of meeting the specific requirement defined in the tree. In that respect, $E_i = 0$ indicates that the performance variable X_i does not meet the predefined requirement, $E_i = 1$ represents the complete satisfaction of the requirement while the partial satisfaction of the requirement is in the range $0 < E_i < 1$. The last step in applying LSP method is aggregation of preferences, a process of calculating the global preference $E_0 = L(E_1, \dots, E_n) = L(G_1(X_1), \dots, G_n(X_n)) = G(X_1, \dots, X_n)$, $0 \leq E_0 \leq 100\%$ (4) that represent the extent to which all requirements in a tree are met. The aggregation function $L: [0,1]^n \rightarrow [0,1]$ (5) is an iterative stepwise process that follows the structure of the requirement tree, going from the leaves towards the root [15]. Theoretically related preferences are aggregated with the appropriate logical operator which results in preference subsystems. The process is continued by aggregating preference subsystems until a single global preference E_0 is calculated. In each step of the aggregation process, it is necessary to select the appropriate logic aggregation operator and determine the relative significance of preferences. Logic aggregation operators are special cases of the generalized conjunction/disjunction function that is implemented by means of the weighted power mean [15].

LSP method has been so far used in evaluation of different types of Web sites. While majority of researchers (e.g. [16][17]) based their assessment approaches on objective metrics, only few of them (e.g. [18]) carried out subjective evaluation by employing a questionnaire as a measuring instrument. Yip and Mendes [19] have found that in the context of measuring usability of Web sites results of objective evaluation based on the employment of LSP significantly differ from subjective assessment. To overcome the aforementioned drawback, the objective of the comprehensive research this work is part of is to develop a method that would, drawing on both subjective and objective metrics, results in composite quantitative indicator with respect to the quality of social Web applications.

4. REQUIREMENT TREE

Quality model is “defined set of characteristics, and of relationships between them, which provides a framework for specifying quality requirements and evaluating quality” [20]. When the quality of social Web applications is considered, the quality model in the form of requirement tree we are proposing is composed of 35 quality attributes. They denote the extent to which particular social Web application: provides various navigation mechanisms (navigability), has uniform interface structure, design, and terminology (consistency), is similar to the previously used applications (familiarity), can be personalized to meet users’ needs (customizability), has implemented mechanisms that protect created and stored artefacts from unauthorized use (security), operates properly on different devices and among various environments (compatibility), can exchange files with other applications and use files that were exchanged (interoperability), provides various forms of help to users (helpfulness), is

continuously reachable (availability), facilitates handling of created artefacts (artefacts management), contains mechanisms that prevent errors to emerge (error prevention), is unailing (reliability), can recover from errors and operational interruptions (recoverability), notifies users with appropriate and useful messages (feedback), supports teamwork and enables different types of communication among users (interactivity), enables users to execute tasks accurately and completely (effectiveness), encourages users to quickly perform tasks (efficiency), responds promptly to users' actions (response time), is capable to operate under an increased or expanding workload (scalability), is usable within and beyond initially intended contexts of use (context coverage), consumes small amount of physical and mental effort when employed (minimal workload), can be used by people with the widest range of characteristics and capabilities (accessibility), provides users a full freedom in executing tasks (controllability), is simple for operation (ease of use), enables users to easily become skilled in interaction with its functionalities (learnability), has easy-to-remember interface features (memorability), is clear and unambiguous (understandability), has visually appealing user interface (aesthetics), is beneficial in the context of executing tasks (usefulness), stimulates users' curiosity and creativity (playfulness), is perceived positively by users (attitude towards use), meets users' expectations (satisfaction), arouses users' emotional responses (pleasure), is distinctive among applications with the same purpose (uniqueness), and encourages users to employ it on regular basis and recommend it to others (loyalty).

5. PERFORMANCE VARIABLES

In the context of evaluating quality of social Web application, we distinguish two types of performance variables: indicators and items. Indicators are performance variables designed for collecting objective data related to estimated facets of quality. They are commonly used when methods such as logging actual use [21] are applied for evaluation purposes. There are six indicators which are relevant for our study. Task completion denotes the proportion of the number of scenario-based tasks particular user completed and the total number of tasks in the scenario. This indicator is designed for evaluating estimated effectiveness. The distance refers to the number of millimeters user travelled while moving the mouse during scenario-based tasks execution. The mouse clicks denote the sum of left, right, and middle mouse clicks for the purpose of performing scenario-based tasks. The mouse wheel scrolls represent the amount of scrolls the user made while reaching the solution of scenario-based tasks. The keystrokes indicate the total number of keys on the keyboard that the user pressed while completing scenario-based tasks. The set forth indicators are intended for measuring estimated workload by means of specialized software such as Mousotron [22]. The time is the amount of seconds required to complete the scenario-based tasks. This indicator is proposed for the assessment of estimated efficiency.

Items are performance variables in the form of questionnaire statements meant for measuring perceived dimensions of quality. Although literature offers a number of questionnaires meant for evaluating software in general (e.g. [23]) and web sites in particular (e.g. [24]), none of them captures all relevant particularities of social Web applications. In addition, they are commonly designed for examining pragmatic aspects of usability or hedonic dimensions of user experience while measuring instruments which combine both facets of quality are rather scarce. In that respect, we developed a post-use questionnaire that enables assessment of all relevant facets of perceived quality that constitute the requirement tree with between three and five items.

6. ELEMENTARY CRITERIA

6.1 Objective Metrics

Determining elementary criteria for six indicators meant for objective assessment of three pragmatic dimensions (effectiveness, workload, and efficiency) of quality in the context of social Web applications was based on findings of our prior studies [25][26].

An elementary criterion for evaluating estimated users' effectiveness in performing predefined scenario steps of interaction with social Web applications designed for collaborative writing and mind mapping are shown in Figure 1. Values on the preference scales were obtained on the basis of the minimum and maximum number of scenario steps that the pilot study participants managed to complete during interaction with the aforementioned types of social Web applications.

The elementary criteria for the remaining five indicators were calculated in several steps. First, values obtained by means of specific indicator were divided by the number of scenario steps that individual participants in the pilot study managed to complete by employing particular social Web application. In that manner, indicator values per scenario step for every participant were calculated. Then, the yielded values were multiplied by the total number of scenario steps in order to calculate the values that the indicators would take if a specific participant succeeded in completing all scenario steps by means of specific social Web application. Finally, by applying the lower and upper interquartile to a sample of indicator values for social Web applications of the same type which the pilot study participants used, minimum and maximum values on the indicators' preference scales were calculated, respectively.

6.1.1 Estimated Effectiveness

Top preference scale presented in Figure 1 indicates that particular social Web application for collaborative writing completely meets requirement of estimated effectiveness ($E_i = 100\%$) if users during an interaction with it can complete all 45 scenario steps (X^{max}) and that the same requirement is not met ($E_i = 0\%$) if users can complete 17 or less scenario steps (X^{min}) in the same respect. On the other hand, bottom preference scale shown in Figure 1 denotes that if users can complete 43 scenario steps (X^{max}) when using social Web application for mind mapping, it completely ($E_i = 100\%$) satisfies requirement of estimated effectiveness but fails to meet this requirement ($E_i = 0\%$) if users can complete 13 or less scenario steps (X^{min}).

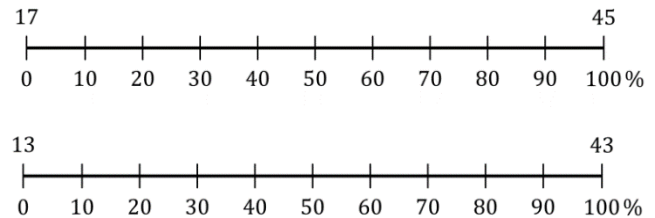


Figure 1. Elementary criteria defined as preference scales for evaluating estimated users' effectiveness in completing tasks by means of social Web applications for collaborative writing (top) and mind mapping (bottom)

For both types of social Web applications, the elementary preference score related to estimated users' effectiveness can be calculated by means of following increasing function composed of three linear parts [14][15]:

$$E_i = G_i(X_i) = \begin{cases} 0, & X \leq X^{min} \\ \max\{0, \min[1, (X - X^{min})/(X^{max} - X^{min})]\}, & X^{min} < X < X^{max} \\ 1, & X \geq X^{max} \end{cases} \quad (6)$$

6.1.2 Estimated Workload

In the context of social Web application for collaborative writing, if users have pressed 591 keyboard keys or less when completing scenario steps, the number of keystrokes pressed as a requirement of estimated workload is considered to be met. On contrary, if the users pressed 1184 keyboard keys or more while performing scenario steps with social Web applications for collaborative writing, this requirement related to estimated workload is not satisfied. When social Web applications for mind mapping are taken into account, number of keyboard keys pressed equal to or less than 642 perfectly meets requirement of estimated workload related to the number of keystrokes pressed whereas number of keyboard keys pressed equal to or greater than 1011 violates the set forth requirement in the context of completing predefined scenario-based tasks.

When the distance as the second requirement of estimated workload is tackled, if users have traveled distance of 75961 millimeters or less while moving the mouse for the purpose of performing predefined scenario steps with social Web application designed for collaborative writing, it has perfectly met this elementary criterion. On the other hand, if users have traveled distance of 169756 millimeters or more in the same respect, social Web application failed to satisfy this requirement of estimated workload. When the aforementioned preference scale is considered in the context of social Web applications for mind mapping, distance traveled of 80834 millimeters or less represents complete fulfillment of the eponymous requirement of estimated workload while distance traveled of 160140 millimeters or more indicates non-compliance with this criterion.

While evaluating estimated workload in interaction with social Web applications for collaborative writing with number of mouse clicks required for completing scenario-based assignments, 484 mouse clicks or less imply perfect compliance with this criterion whereas 1038 mouse clicks or more represent violence of this requirement in the context of estimated workload assessment. When evaluating social Web applications for mind mapping in the same respect, 534 mouse clicks denote that this requirement of estimated workload is completely satisfied while 991 mouse clicks or more signify that this requirement of estimated workload has not been met.

The number of mouse wheel scrolls equal or less than 130 indicates that eponymous requirement of estimated workload is met with regard of completing scenario-based tasks by employing social Web application designed for collaborative writing. As opposed to that, 871 or more mouse wheel scrolls instruct that collaborative editor failed to comply with this requirement of estimated workload. In the context of social Web applications for mind mapping, 22 mouse wheel scrolls or less imply perfect compliance with this elementary criterion while 450 or more mouse wheel scrolls signify the violence of the eponymous requirement.

6.1.3 Estimated Efficiency

When examining estimated efficiency with respect to social Web applications for collaborative writing, time of 2506 seconds or shorter means perfect fulfillment of the eponymous requirement while time of 5900 seconds or longer uncovers non-compliance with this criterion. In the context of social Web applications for mind mapping, time of 2466 seconds or shorter implies complete

satisfaction of this requirement of estimated efficiency while time of 5192 seconds or longer represents violence of this requirement.

Values of elementary preferences for the aforementioned five indicators (number of keystrokes, number of mouse clicks, distance traveled while moving the mouse, number of mouse wheel scrolls, and time required for completing the assignments), can be calculated with following decreasing function composed of three linear segments [14][15]:

$$E_i = G_i(X_i) = \begin{cases} 1, & X \leq X^{min} \\ \max\{0, \min[1, (X^{max} - X)/(X^{max} - X^{min})]\}, & X^{min} < X < X^{max} \\ 0, & X \geq X^{max} \end{cases} \quad (7)$$

6.2 Subjective Metrics

The elementary criterion for subjective quality evaluation by means of five-point Likert scale of five degrees (1 - completely agree, 5 - completely disagree) is shown in Figure 2.

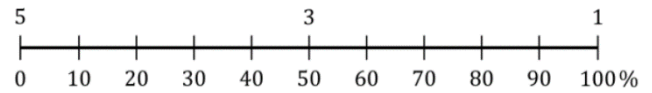


Figure 2. Elementary criterion defined as preference scale for evaluating perceived quality of social Web applications

Unlike the objective elementary criteria that differ depending on the employed indicator, the subjective elementary criteria are the same for all quality evaluation items. Value 1 on the Likert scale indicates that particular subjective requirement is perfectly met, value 3 implies that 50% of the subjective requirement is satisfied, while value 5 denotes that subjective requirement has not been met. The remaining two values are calculated by linear interpolation. The elementary preference for subjective quality evaluation is calculated using the following function:

$$E_i = G_i(X_i) = \begin{cases} 1, & X \leq X^{min} \\ 0.25(5 - X), & X^{min} < X < X^{max} \\ 0, & X \geq X^{max} \end{cases} \quad (8)$$

7. CONCLUSION

The work presented in this paper is a part of the comprehensive research whose objective is to develop a methodology that would facilitate comparison of evaluated social Web applications as well as their assessment with respect to all users' requirements in the quality model. As one of the first steps towards this goal, objective and subjective metrics have been proposed. Although reference values presented in this paper can be generalized only to social Web applications meant for collaborative writing and mind mapping which were case study in our research, the manner of determining them can be applied to all types of social Web applications. Considering that we already involved field experts for the purpose of identifying the relevance of quality attributes [27], our future work will be focused on determining significance of preferences in the requirement tree, selecting the appropriate logical operators for aggregating preference subsystems in order to calculate a single global preference that would reflect an overall quality of social Web applications.

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